Hospital Chatbot for Color-Coded Clinical Pathways

Synopsis Presentation

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Contents

1. Introduction (Background of Study)

"Triage is not just about sorting; it is about the right patient getting the right care at the right time."
– South African Triage Group

- In high-functioning systems, triage is a **continuous, dynamic status** that dictates the patient's entire hospital journey.
- Standardized scales use discrete color codes to classify clinical severity. Common scales include:
  - **SATS (South African Triage Scale):** A widely used standard for assessing both adults and older children [1].
  - **ETAT (Emergency Triage Assessment and Treatment):** Guidelines specifically designed by the WHO for triaging sick children [2].
- These colors are not just labels; they are **triggers for specific Clinical Pathways** with strict time limits.

1. Introduction: The Triage Color Codes

Standardized Clinical Severity

- **Red**: Emergency (Immediate life-saving care needed)
- **Orange**: Very Urgent (Care needed within 10 minutes)
- **Yellow**: Urgent (Care needed within 60 minutes)
- **Green**: Non-Urgent (Can wait safely or be redirected)
- **Blue**: Dead (Deceased on arrival)

The LMIC Challenge: In many resource-constrained settings, the utility of this color often ends at the front desk. The hospital "forgets" the priority level once the patient leaves the triage area [4].

2. Problem Statement: Systemic Bottlenecks

Current hospital workflows suffer from a critical failure known as **Acuity Discontinuity**.

What is Acuity Discontinuity?

It occurs when the hospital "forgets" how sick a patient is. A patient might be marked as high-priority at the front door, but they lose that priority and join a normal queue when transferred to another department.

This creates four major problems:

1. **Long Waiting Lines:** Patients get sicker while waiting just to be seen.
2. **Human Error:** Nurses manually computing the **Triage Early Warning Score (TEWS)** in their heads make subjective mathematical mistakes.
3. **Hidden Deterioration:** Waiting patients get worse, but the system doesn't notice.
4. **Lost Information:** Priority levels aren't shared across departments.

3. Objectives

General Objective

To design a robust, logic-based digital framework and computational engine that automates the continuous management of clinical pathways. This core routing system must be built and functioning **first**, with a chatbot later serving as the user-friendly frontend interface.

3. Specific Objectives

- **1. Automate Acuity Computation (Core Engine):** Build a mathematical engine to objectively calculate the TEWS danger scores, entirely removing human math errors.
- **2. Connected Routing:** Define strict logistical rules so the priority color remains digitally attached to the patient across all hospital departments.
- **3. Watchful Timers:** Engineer digital stopwatches to alert staff to re-check patients who wait too long.
- **4. Fast Check-in Interface:** *After the engine is built*, design a chatbot frontend to quickly collect patient symptoms upon arrival.
- **5. Test Success:** Use computer simulation models to prove this automated routing system is faster than manual paperwork.

4. Methodology: The Computational Engine (Phase 1)

The foundation of this framework is a **universally integratable System Architecture**. The triage process is formally modeled as a robust mathematical engine, independent of any specific frontend.

Step 1: The Automatic Danger Score (TEWS)
An integrated frontend interface (such as our chatbot) extracts a vector of vital signs $V_i = [HR, RR, SBP, T, AVPU]$. The universal backend engine evaluates these using a weighted summation function:

$$TEWS(P_i) = \sum_{k=1}^{n} w_k \cdot f_k(v_k)$$

Detailed Breakdown of the Equation:

- $TEWS(P_i)$**:** The final Danger Score calculated for a specific Patient (P_i).
- v_k**:** The raw value of a specific vital sign (e.g., Heart Rate of 120 bpm).
- f_k**:** A mathematical function that measures how far that specific vital sign (v_k) deviates from a normal, healthy baseline.
- w_k**:** The clinical weight or importance of that vital sign.
- $\sum_{k=1}^{n}$: The system adds up these calculations for all n vital signs collected.

4. Methodology: Instant Color Sorting

Step 2: Assigning the Clinical Color

The system maps the computed TEWS score into a discrete clinical color using a strict piecewise function, entirely removing human bias:

$$C(TEWS) = \begin{cases} \textbf{Red (Emergency)} & \text{if } TEWS > 7 \\ \textbf{Orange (Very Urgent)} & \text{if } 5 \leq TEWS \leq 6 \\ \textbf{Yellow (Urgent)} & \text{if } 3 \leq TEWS \leq 4 \\ \textbf{Green (Non-Urgent)} & \text{if } TEWS \leq 2 \end{cases}$$

Detailed Breakdown:

- $C(TEWS)$: The Category or Color assigned based on the Danger Score.
- **Logic Constraints** ($\leq, >$): These strict numerical boundaries act as an automated digital filter. If a score is 5, it cannot be subjectively argued as Yellow; the math strictly categorizes it as Orange.

4. Methodology: Digital GPS & Routing (Phase 2)

Once the color is set, the system acts like a hospital GPS. It strictly controls where the patient is allowed to go based on their color:

- **Red Pathway:** The system forces a direct route to the emergency room, skipping all registration and payment desks to save time.
- **Yellow Pathway:** The system sends the patient to the waiting room but simultaneously sends automatic digital requests to the Lab to prepare tests in advance.
- **Green Pathway:** The system redirects healthy, non-urgent cases away from the ER and into regular outpatient clinics to prevent overcrowding.

4. Methodology: Timers & Validation (Phases 3 & 4)

Phase 3: The Digital Stopwatch

- The system constantly tracks elapsed waiting time:

$$\Delta t = t_{current} - t_0$$

  - Δt**:** The total time the patient has spent waiting.
  - $t_{current}$**:** The actual current time on the system's clock.
  - t_0**:** The exact timestamp when the patient was initially triaged.
- If Δt approaches their safety limit (T_{max} e.g., 60 mins for Yellow), an alarm triggers.
- This forces staff to collect new vitals. The engine instantly **recalculates the TEWS**, allowing for dynamic color upgrades if they worsen.

Phase 4: Integration & Validation

- **Universal Integration:** The chatbot serves as just one example of an interactive terminal that can feed data into this scalable mathematical backend.
- **Proving it Works:** System efficiency is validated via retrospective computer simulation using historical hospital data.

5. Analysis and Expected Results

Evaluation Metrics

- **Sensitivity:** Did the TEWS algorithm catch all truly sick patients?
- **Specificity:** Did it correctly filter out the healthy ones?
- **Process Efficiency:** Reduction in total wait times [4].

Expected Outcomes

- A 24/7 "Digital Architect" free from computational fatigue.
- Significant reduction in ED congestion via Green Path diversions.
- Systemic safety through mathematically guaranteed routing.

6. Conclusion & Recommendations

Conclusion

- Transforming triage from a manual task to a mathematically-driven navigation system addresses the root causes of overcrowding.
- Rigid algorithms ensure safety is *systemic*, not accidental.
- The chatbot is the user-friendly frontend to a computational engine that saves lives and optimizes flow.

Recommendations

- LMIC hospitals should adopt digital tools to *support*, not replace, clinical staff [2].
- Future policy should explicitly integrate "Blue Pathways" for structured efficiency.

References

[1] Rominski, S., et al. (2014). The implementation of the South African Triage Score (SATS) in an urban teaching hospital, Ghana. *African Journal of Emergency Medicine*.

[2] Kamau, S., et al. (2024). Comparison between the Smart Triage model and ETAT guidelines in triaging children in Kenya. *PLOS Digital Health*.

[3] Nurchis, M., et al. (2025). Assessing the Impact of Waiting Time on Triage Color Code Assignment and One-Year Mortality. *Health Science Reports*.

[4] Mitchell, R., et al. (2023). Systematic review: What is the impact of triage implementation on clinical outcomes in LMIC emergency departments? *Academic Emergency Medicine*.